

Tianruo Rose Xu

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Education

Cornell University, College of Engineering
B.S. in Computer Science Minor in Mathematics

Aug 2024 – May 2027

Honors and Awards

- Hunter R. Rawlings III Cornell Presidential Research Scholar; Clare Boothe Luce Undergraduate Research Award.
- Menschel Cornell Commitment Public Service Grant; Cornell Graduate School Opportunity Fund Grant.
- Bowers Undergraduate Research Experience (BURE) stipend and Academic Year Expansion support.
- USA Computing Olympiad (USACO) Platinum Division; full score in the [January 2023 Gold Contest](#).
- USA Mathematical Olympiad (USAMO) Qualifier.

Publications

[Preprint] **Tianruo Rose Xu**, Jiawei Ren, Yichi Yang, Zhaoxu Zheng, Lianhui Qin. *RT-Safe: Benchmarking Agent Safety in Real-Time Embodied Environment*. arXiv preprint arXiv:2610.09294, October 2026. [\[Paper\]](#) [\[Project Website\]](#) [\[Code\]](#).

[Preprint] Haoqiang Kang, Yiming Zhang, Yiyang Guo, Chuying Li, Jianzhi Shen, **Tianruo Rose Xu**, Xiaokang Ye, Lianhui Qin. *DeliveryGym: An RL Environment for Long-Horizon Embodied Agent Planning with Adaptive Curriculum*. arXiv preprint arXiv:2609.19801, September 2026. [\[Paper\]](#) [\[Code\]](#).

[Submitted] *Can Emergent Verbalization of Epistemic Uncertainty Inform Factuality?*

[Published] **Tianruo Rose Xu**. *Can You Trust What I Think? Analyzing and Improving Verbalized Uncertainty and Factuality in Reasoning-Based Large Language Models*. AAAI-26 Undergraduate Consortium, Singapore, January 2026. [\[Poster\]](#) [\[Oral Presentation\]](#) [\[Paper\]](#).

[Published] **Tianruo Rose Xu**, Vedant Gaur, Liu Leqi, Tanya Goyal. *The Progress Illusion: Revisiting Meta-Evaluation Standards of LLM Evaluators*. Findings of the Association for Computational Linguistics: EMNLP 2025, Suzhou, China, November 2025. [\[Paper\]](#) [\[Poster\]](#).

[Published] Yingke Ding, Jiankai Tang, Wanying Mo, **Tianruo Rose Xu**, Yuanchun Shi, Yuntao Wang. *NFCMTL: Auto NailFold Capillaroscopy through a Multi-Task Learning Model*. MICCAI 2025 Workshop MSB EMERGE, July 21, 2025. [\[Paper\]](#) [\[Code\]](#). Runner-up Best Paper, MICCAI 2025 Workshop MSB EMERGE.

[Published] Tianruo Xu. *Elder Depression Detection by Multimodal Means*. BCP Education & Psychology, November 7, 2022. [\[Paper\]](#).

Research Experience

Undergraduate Researcher, Cornell University

Computer Science Department, NLP Group; advised by Professor Tanya Goyal and Claire Cardie.

Analyzing Verbalized Uncertainty and Factuality in Reasoning-Based LLMs 2025 – 2026

- Investigate whether epistemic uncertainty verbalized in LLM reasoning predicts fact-level factuality in long-form QA.
- Led project design with Anirudh Atmakuru and Sanjana Ramprasad; built pipelines for atomic fact extraction, trace-response alignment, FactScore evaluation, uncertainty analysis, and classifier training.

Revisiting Meta-Evaluation Standards of LLM Evaluators 2025

- Led fine-grained meta-evaluation of LLM judges: designed evaluation frameworks, built data and scoring pipelines, and analyzed evaluator reliability, reference-model sensitivity, and correlations with human judgments.

Content and Style Evaluation in Natural Language Generation Systems 2024

- Designed experiments and evaluation pipelines to study content and style as separate dimensions of generation quality; conducted model-based evaluations, error analysis, ablations, and result visualization.

Research Experience (continued)

Research Intern, UC San Diego

Computer Science Department, Q-Lab; advised by Professor Lianhui Qin.

RT-SAFE: Real-Time Embodied Agent Safety **2026**

- Lead RT-SAFE, a benchmark evaluating embodied-agent safety while environments evolve during inference and action.
- Defined the research direction, safety scenarios, tasks, and evaluation protocol; led analysis of how inference latency and reasoning effort affect navigation safety.

DeliveryGym / DeliveryBench: Embodied Agent Environments **2026**

- Built traffic, obstacle, and safety infrastructure for VAGEN/DeliveryBench, including pedestrian signals, state checks, hazard evaluation, and the Unreal Engine first-person observation pipeline.
- Integrated the CityCore Paris map with semantic points of interest, buildings, navigation graphs, coordinate transforms, and visual assets; developed traffic-signal and blocked-road variants.

Undergraduate Researcher, Cornell University

Mathematics Department, Math + AI Lab; advised by Professor Daniel Halpern-Leistner.

Lean Proof Arena and LaTeX-to-Lean Autoformalization Pipeline **2026**

- Develop Lean Proof Arena and a LaTeX-to-Lean pipeline for DARPA ExPMath-supported research-level autoformalization; build benchmarks and evaluation protocols for formal correctness and semantic fidelity.

Undergraduate Research Assistant, Tsinghua University

Department of Computer Science and Technology, Pervasive Human-Computer Interaction Lab.

Intelligent AI Tool Benchmarking and Orchestration Platform **2025 – 2026**

- Designed back-end search and matching algorithms for an AI tool platform supporting standardized capability descriptions, user-intent matching, personalized tool selection, and multi-tool orchestration.

NFCMTL: Auto NailFold Capillaroscopy through a Multi-Task Learning Model **2025**

- Preprocessed data, trained baselines, and evaluated a multi-task model combining segmentation, classification, and keypoint detection for automated nailfold capillaroscopy.

Undergraduate Research Assistant, Cornell University **2024**

Information Science Department; advised by Keigo Kusumegi and Professor Yian Yin.

- Processed Wikipedia data and applied K-means clustering to study collaboration patterns and article quality.

Teaching Experience

Cornell University

- **Head Teaching Assistant**, CS 3780: Introduction to Machine Learning Fall 2026
- **Climbing Instructor**, PE 1640: Basic Rock Climbing Fall 2026
- **Teaching Assistant**, CS 3780: Introduction to Machine Learning Spring 2026
Designed the final Kaggle challenge, [All About Birds!](#), a bird-sound spectrogram classification task.
- **Teaching Assistant**, CS 4820: Introduction to Analysis of Algorithms Fall 2025
- **Teaching Assistant**, CS 2800: Mathematical Foundations of Computing Spring 2025

Industry Experience

Incoming Machine Learning Engineer Intern, TikTok **2027**

TikTok E-commerce Knowledge Graph; NLP/CV/Multimodal/Algorithm. Seattle, WA

Independent Research Collaboration, Oracle **2025 – 2026**

Remote; mentored by Priyan Pattnayak, Senior Principal Scientist at Oracle Cloud.

- Received mentorship on the AAAI publication.
- Currently investigating vision-language models for sign language understanding.